# Collaborative Filtering for MeSH Keyword Recommendation on Clinical Trial Records

UE24CS352A · Machine Learning mini-project · Ramneek Gautam and Rufus Thomas · Technical report

## Executive summary

This project builds a keyword recommender for clinical-trial search. Given the MeSH concepts detected in a short query, the system predicts additional concepts that may help broaden retrieval. Trials are represented as documents, MeSH terms as items, and TF–IDF values as the utility signal. The existing presentation reports top-3 precision (APTK) of **0.4776** for CF + ReLU, compared with 0.4476 for popularity, 0.4162 for item-item kNN, and 0.0152 for a random ranking. These are retained as the reference-slide results, not silently replaced by results from a separate later run.

## 1. Problem statement

Short clinical-trial queries can omit relevant medical terminology. Lexical retrieval systems cannot consistently recover concepts that are absent from the query representation. The recommender is designed to predict likely missing MeSH concepts so they can be appended to a query before retrieving similar trials. The task is query expansion, not diagnosis or treatment recommendation.

## 2. Dataset and representation

The source write-up describes 50,000 ClinicalTrials.gov records and the 2026 MeSH descriptor file. Text fields include brief and official titles, brief summaries, detailed descriptions, conditions, and eligibility criteria. Entry-term synonyms are mapped to canonical concepts using a word-level Trie with longest-match-first extraction. The utility matrix contains 50,000 documents and 1,000 frequent MeSH terms; documents are randomly split 80/10/10 into training, development, and test sets.

The reported TF–IDF utility is U■■ = log■■(1 + count■■) × log■■(N / df■), which is non-negative and exactly zero when a term is absent. The source write-up reports 3.2% non-zero cells and excludes the ambiguous concept “Volition”. Matrix density is run-dependent in the later demo logs; density values should be attached to their respective run metadata rather than merged.

## 3. Model

For term i in trial j, the model estimates its TF–IDF value from neighbouring term values: $R■■■ = ReLU(\Sigma f\ W■f\ \Theta■f\ Rf■ + b■)$. The neighbour matrix W is fixed from cosine similarity between term profiles, with a zero diagonal to prevent direct self-prediction. Θ and b are learned by mini-batch gradient descent, with an L2 penalty on Θ. ReLU constrains estimates to be non-negative. The implementation is described as a NumPy implementation with hand-derived gradients.

## 4. Evaluation protocol and baselines

For each test document, 50% of non-zero terms are masked. The model ranks candidate terms and APTK (K=3) measures average precision among the top three recommendations relative to the masked terms. APRK is the random-ranking reference. MSE and MSER quantify reconstruction error. The displayed baselines are a popularity recommender and item-item cosine kNN with k=50. The source slide notes that the result is not directly comparable with the paper’s 0.50 because the paper used 15,071 terms, while this implementation uses 1,000.

## 5. Results retained from the existing presentation

| Model | APTK (K=3) | Interpretation |
|---|---|---|
| Random | 0.0152 | Reference floor |

| Model | APTK (K=3) | Interpretation |
|---|---|---|
| Popularity | 0.4476 | Strong frequency baseline |
| Item-item kNN (k=50) | 0.4162 | Neighbour-based baseline |
| **CF + ReLU** | **0.4776** | **Highest score among displayed models** |

From these retained figures, CF + ReLU exceeds popularity by 0.0300 absolute APTK (3.0 percentage points) and item-item kNN by 0.0614 (6.1 points when rounded to one decimal point). The result establishes the ranking order on the reported evaluation, but does not alone establish better end-to-end retrieval.

## 6. Qualitative behaviour and limitations

The accompanying write-up reports that hand-written queries about lung cancer, hypertension, and Alzheimer's disease often received similar generic terms, including Therapeutics, Safety, Disease, Child, and General Surgery. A raw-score ranking can favour concepts whose predicted values are high across many documents. A query-specific score that subtracts a typical prediction was added as an exploratory demo option, but it was not evaluated with APTK in the source report. Its effect on ranking quality remains unverified.

Other limitations include a single random train/test split; only 1,000 terms because of the quadratic cost of term-to-term similarities; lower-cased matching that can produce plain-language false positives; and evaluation on masked-term recovery rather than end-to-end retrieval. Bootstrap intervals quantify sampling uncertainty for the held-out documents on that split, not variability across alternative splits.

## 7. Experimental integrity and next steps

Hyperparameter selection should use the development set. The test set should remain untouched until the selected configuration is frozen. Experiments using different vocabularies, matrix caches, masks, or splits must be labelled as separate runs. The later demo run visible in the project conversation reports APTK 0.3193 for CF + ReLU, 0.3016 for item-kNN, 0.1064 for popularity, and 0.0127 for random. Those values differ from the existing slide's 0.4776 / 0.4162 / 0.1064 / 0.0127 comparison and must not be substituted or compared as though they came from an identical protocol.

### Recommended next experiments

1. Audit cache invalidation and ensure each vocabulary cutoff rebuilds the matrix and split consistently. 2. Select hyperparameters on development APTK and retain the complete search log. 3. Evaluate the frozen configuration once on the held-out test set. 4. Score the query-specific ranking with the same held-out protocol. 5. Measure retrieval quality with and without recommended terms, rather than relying only on masked-term precision.

## Conclusion

The existing results support CF + ReLU as the best of the displayed models on the reported 1,000-term APTK comparison. The next research question is whether that advantage survives stricter, reproducible comparisons and translates into more relevant clinical-trial retrieval. The current evidence supports a promising prototype, not a validated clinical search system.

Source basis: CF Keyword Recommender for Clinical Trials presentation and Writeup_CF_MeSH_Recommender (2).pdf; PEEVED_Mini_Project_Review_Final.pptx used as a reference for analytical structure and presentation depth.